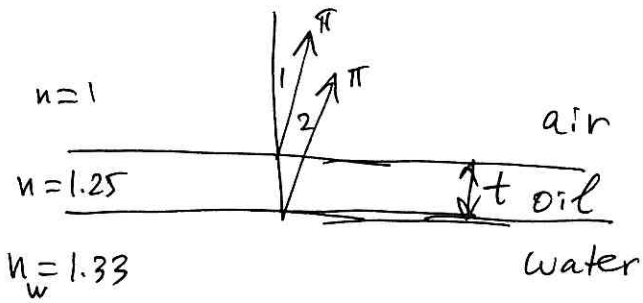
HW02

P1



both rays are π-shifted ⇒ this does not affect path difference.

$$\delta_{red} = n\,k_{red} \cdot 2t = 2\pi m_1 \quad \text{constructive}$$

$$\delta_{green} = n\,k_{green} \cdot 2t = (2m_2+1)\pi \quad \text{destructive}$$

$$\Rightarrow 2t = \frac{2\pi m_1}{n\,k_{red}} = \frac{(2m_2+1)\pi}{n\,k_{green}}$$

$$k = \frac{2\pi}{\lambda}$$

$$\frac{\lambda_{red}\, m_1}{n} = \frac{(m_2+\frac{1}{2})\lambda_{green}}{n}$$

$$1.25 = \frac{5}{4} = \frac{640}{512} = \frac{\lambda_{red}}{\lambda_{green}} = \frac{m_2+\frac{1}{2}}{m_1}$$

$5m_1 = 4(m_2+\frac{1}{2})$ ⇒ minimal integer solution

$m_1 = 2,\ m_2 = 2$

$$\text{Then } 2t = \frac{2\pi m_1}{n\,\frac{2\pi}{\lambda_{red}}} = \frac{m_1\lambda_{red}}{n}$$

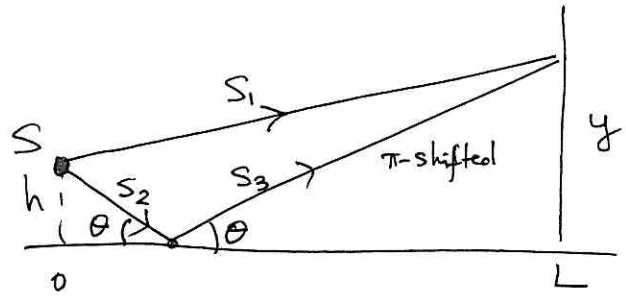
$$t = \frac{2\cdot 640\,nm}{2\cdot 1.25} = 512\,nm = \lambda_{green}\ !$$

P2

$\lambda = 500\,nm$

$h = 0.01\,m$

$L = 100\,m$



dark band: $\delta = (2m+1)\pi$.

$\delta = K(S_2 + S_3 - S_1) + \pi$

↳ due to reflection from the mirror

$$s_1 = \sqrt{L^2 + (y-h)^2} = L\sqrt{1 + \left(\frac{y-h}{L}\right)^2} \approx L + \frac{(y-h)^2}{2L}.$$

$$\left.\begin{aligned} s_2 \sin\theta &= h \\ s_3 \sin\theta &= y \end{aligned}\right\} \quad s_2 \cos\theta + s_3 \cos\theta = L$$

$$\Rightarrow (h+y)\frac{\cos\theta}{\sin\theta} = L, \quad \tan\theta = \frac{h+y}{L}$$

But $h+y \ll L \Rightarrow \tan\theta \approx \sin\theta \approx \theta = \frac{h+y}{L}$.

Hence $s_2 + s_3 = \frac{L}{\cos\theta} \cong \frac{L}{1-\frac{1}{2}\theta^2} = L\left(1 + \frac{(h+y)^2}{2L^2}\right)$

$$\Rightarrow \delta = \frac{2\pi}{\lambda}\left(L + \frac{(h+y)^2}{2L} - L - \frac{(h-y)^2}{2L}\right) + \pi =$$

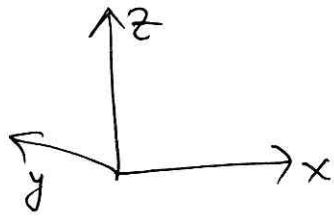
$$= \frac{2\pi}{\lambda}\,\frac{4hy}{2L} + \pi = (2m+1)\pi$$

$$\Rightarrow \frac{2\pi}{\lambda}\,\frac{2hy}{L} = 2\pi m \Rightarrow y = m\frac{\lambda L}{2h}$$

First dark band $m=1$: $y = \frac{500 \cdot 10^{-9}\,m \cdot 10^{2}\,m}{2 \cdot 10^{-2}\,m} = 2.5 \cdot 10^{-3}\,m$

$= 0.25\,cm$

P3



$$E_{10} = E_{20} \cdot 2$$

$$\vec{E}_1 = E_{10}\,\hat{y}\,\cos(\omega t - kx)$$

$$\vec{E}_2 = E_{20}\,\hat{z}\,\cos(\omega t - kx + \tfrac{\pi}{2}) = -E_{20}\,\hat{z}\,\sin(\omega t - kx)$$

$$\vec{E} = \vec{E}_1 + \vec{E}_2 = E_{10}\cos(\omega t - kx)\,\hat{y} - E_{20}\sin(\omega t - kx)\,\hat{z}.$$

This is elliptically polarized light as

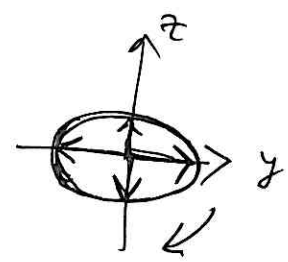
$$\left(\frac{E_y}{E_{10}}\right)^2 + \left(\frac{E_z}{E_{20}}\right)^2 = 1 \quad \text{(ellipse equation)}$$

or $E_y^2 + 4E_z^2 = E_{20}^2$.

Sense of rotation? Take $x=0$:

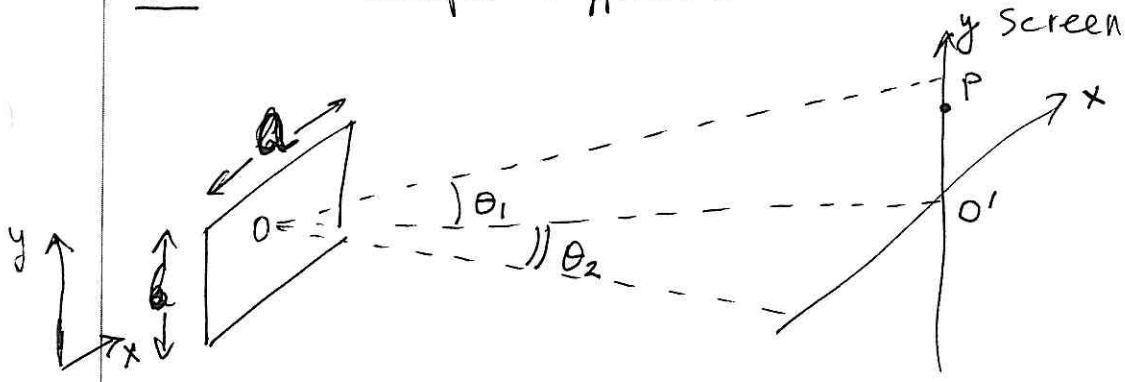
at $t=0$ $\quad \vec{E} = E_{10}\,\hat{y}$

$t = \frac{\pi}{2\omega}$ $\quad \vec{E} = -E_{20}\,\hat{z}$



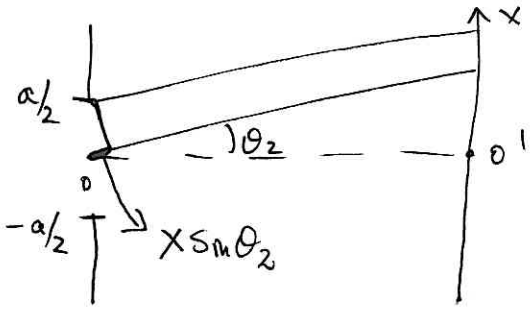
$\Rightarrow$ clockwise rotation (from positive $\hat{y}$ to negative $\hat{z}$)

P4 Fraunhofer diffraction

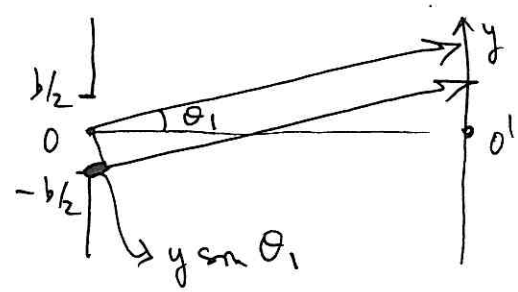


$$E_p \sim \int_{\text{aperture}} e^{ikr}\, dA$$

Projection on $OO'x$ plane:



Projection on $OO'y$ plane



$$kr = k\left(r_0 + x\sin\theta_2 + y\sin\theta_1\right)$$

$$E_p \sim e^{ikr_0} \int e^{ikx\sin\theta_2}\, e^{iky\sin\theta_1}\, dx\, dy =$$

$$= e^{ikr_0} \int_{-a/2}^{a/2} e^{ikx\sin\theta_2}\, dx \int_{-b/2}^{b/2} e^{iky\sin\theta_1}\, dy =$$

$$= e^{ikr_0} \cdot \left.\frac{e^{ikx\sin\theta_2}}{ik\sin\theta_2}\right|_{-a/2}^{a/2} \cdot \left.\frac{e^{iky\sin\theta_1}}{ik\sin\theta_1}\right|_{-b/2}^{b/2} =$$

$$= e^{ikr_0}\, \frac{2i\sin\left(\frac{1}{2}ka\sin\theta_2\right)}{ik\sin\theta_2} \cdot \frac{2i\sin\left(\frac{1}{2}kb\sin\theta_1\right)}{ik\sin\theta_1} =$$

$$= e^{ikr_0}\, \underbrace{ab}_{\text{area}}\, \frac{\sin\left(\frac{1}{2}ka\sin\theta_2\right)}{\frac{1}{2}ka\sin\theta_2} \cdot \frac{\sin\left(\frac{1}{2}kb\sin\theta_1\right)}{\frac{1}{2}kb\sin\theta_1}$$

$$\text{Intensity} \sim |E_p|^2 \sim \underbrace{\left(\frac{\sin\left(\frac{\pi a \sin\theta_2}{\lambda}\right)}{\frac{\pi a \sin\theta_2}{\lambda}}\right)^2}_{\text{Fraunhofer pattern along } x \text{ axis}} \cdot \underbrace{\left(\frac{\sin\frac{\pi b \sin\theta_1}{\lambda}}{\frac{\pi b \sin\theta_1}{\lambda}}\right)^2}_{\text{along } y\text{-axis}}$$

$$k = \frac{2\pi}{\lambda}$$

$$I = I_0 \left(\frac{\sin\alpha}{\alpha}\right)^2 \cdot \left(\frac{\sin\beta}{\beta}\right)^2$$

$$\alpha \equiv \frac{\pi a \sin\theta_2}{\lambda}, \quad \beta = \frac{\pi b \sin\theta_1}{\lambda}$$

Main Maximum for $\alpha = 0, \beta = 0$

First minimum at $\alpha = \pi, \beta = \pi$

Result = product of two one-dimensional patterns

* The result can also be obtained by explicit summation, similar to 1d slit done in the class. But that takes more time...